

Combining Genomics With Lipid and Inflammatory Biomarkers to Predict Coronary Artery Disease Risk

UK Biobank Study

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ABSTRACT

BACKGROUND Coronary artery disease (CAD) polygenic risk score (PRS), low-density-lipoprotein cholesterol (LDL-C), lipoprotein(a) (Lp(a)), and high-sensitivity C-reactive protein (hsCRP) are biomarkers that predict CAD. It is unclear whether integrating genomics with lipid and inflammatory biomarkers could complement traditional risk scores in identifying people at risk of CAD.

OBJECTIVES This study assesses the predictive value of CAD PRS, LDL-C, Lp(a), and hsCRP for incident CAD across different age and sex groups.

METHODS Participants (n = 215,695) from the UK Biobank aged 40 to 69 years with baseline CAD PRS, LDL-C, Lp(a), and hsCRP values were followed for 12 years to assess the incidence of CAD. We evaluated a multivariable-adjusted Cox model that included all 4 biomarkers, net reclassification index, C-statistics, and population attributable risk across different age and sex groups.

RESULTS Over a 12-year follow-up, 4,721 men and 2,425 women developed CAD. The HRs for incident CAD associated with each biomarker elevation were 1.79 (95% CI: 1.70-1.89) for CAD PRS, 1.60 (95% CI: 1.48-1.66) for LDL-C, 1.20 (95% CI: 1.12-1.29) for Lp(a), and 1.64 (95% CI: 1.57-1.72) for hsCRP. CAD PRS demonstrated a stronger association in men (HR per SD: 1.49; 95% CI: 1.45-1.54) than women (HR per SD: 1.37; 95% CI: 1.31-1.44; *P*-interaction ≤ 0.001). All biomarkers conferred greater HRs at younger ages (*P* < 0.0001). Individuals with all biomarkers elevated had a 4.65-fold increased risk of CAD compared with those with no elevated biomarkers. A combined 4-biomarker model had a higher C-statistic of 0.753 compared with the pooled cohort equations (C-statistic of 0.740). The C-statistic of the combined 4-biomarker model was also higher in younger individuals in both sexes and yielded a 32.0% continuous net reclassification index when compared with the pooled cohort equations.

CONCLUSIONS CAD PRS, LDL-C, hsCRP, and Lp(a) show independent age- and sex-specific associations with CAD. Measuring all 4 biomarkers may improve midlife CAD risk prediction for both male and female patients. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS
AND ACRONYMS****CAD** = coronary artery disease**CAD PRS** = coronary artery disease polygenic risk score**hsCRP** = high-sensitivity C-reactive protein**LDL-C** = low-density-lipoprotein cholesterol**Lp(a)** = lipoprotein(a)**NRI** = net reclassification index**PAR** = population attributable risk**PCE** = pooled cohort equations**UKB** = UK Biobank

Cardiovascular disease remains the leading cause of mortality globally, and early identification of high-risk individuals for targeted prevention strategies is critical.¹ Blood-based biomarkers such as lipoprotein(a) (Lp(a)), low-density-lipoprotein cholesterol (LDL-C), and high-sensitivity C-reactive protein (hsCRP) levels demonstrate robust predictive value of 5-year, 10-year, and 30-year risk of cardiovascular disease, providing an opportunity to identify people with a single blood test that could otherwise be missed in contemporary prevention that relies on clinical risk calculators.^{2,3}

In addition to their predictive value, the 3 biomarkers have separate modifiable mechanisms of coronary artery disease (CAD) pathophysiology (ie, lipid, inflammation) that inform differential therapeutic strategies. In contemporary preventive paradigms, residual cholesterol and inflammatory risk are crucial in personalizing preventive therapies after statins.^{4,5} Randomized placebo-controlled trials that safely reduce inflammation have been shown to reduce major adverse cardiovascular events, as they target independent modifiable pathways of CAD pathophysiology.⁶⁻⁸ Phase III trials evaluating the impact of reducing Lp(a) on major adverse cardiovascular events are ongoing.^{9,10} The European Atherosclerosis Society and European Society of Cardiology have endorsed 1-time Lp(a) screening for primary prevention.^{11,12} Adjunct screening of residual inflammatory risk with hsCRP has also been proposed in American guidelines and is recently gaining traction with approval of low-dose colchicine to lower the inflammatory risk of CAD.¹³⁻¹⁶

Although blood biomarkers have shown potential for predicting cardiovascular events, CAD polygenic risk score (CAD PRS), a measure of genetic predisposition to disease, enables earlier risk prediction for CAD across the life course, with improved discriminative capabilities over clinical risk in younger individuals.¹⁷⁻¹⁹ CAD PRS is an independent risk factor for CAD, with a contribution greater than that of many established blood biomarkers and clinical risk factors.^{18,20-22} Given that CAD is a highly heritable disease,^{23,24} PRS offers unique value by enabling early-life risk stratification well before traditional risk factors.

To date, most work has examined lipids, inflammation, and genetics in parallel, whereas our study seeks to integrate inherited and modifiable biology into a single age- and sex-specific framework for CAD

WHAT CLINICIANS NEED TO KNOW

A single measurement of 4 blood-based biomarkers in midlife, namely CAD PRS, hsCRP, LDL-C, and Lp(a), is strongly predictive of future CAD incidence in both sexes and across the age spectrum from 40 to 69 years. A biomarker-based prediction model achieves comparable performance with clinical risk calculators and may identify populations who are not detected using a traditional clinical risk calculator, creating an opportunity for improved primary prevention, particularly in younger age groups, where the performance of biomarker-based screening is highest. Interpreting risk as a cumulative burden of genetic, lipid, and inflammatory pathways provides a straightforward approach to prioritizing earlier, targeted primary prevention.

risk prediction. A combined prediction model incorporating both inherited and modifiable factors is likely to provide the strongest predictive capacity for CAD. Prior work has established the value of combined prediction using CAD PRS and clinical risk calculators, particularly in early middle age or younger.^{17,25-27} More recently, a single biomarker measurement of LDL-C, hsCRP, and Lp(a) has been shown to predict long-term major adverse cardiovascular events.^{2,3} Although prior studies have examined these markers individually, limited work has integrated all 4 major causal domains of CAD, genetic susceptibility with CAD PRS, lipoprotein biology with LDL-C and Lp(a), and vascular inflammation through hsCRP into a single predictive biological framework for CAD risk prediction.

This integrated, biologically driven framework is now possible and timely because of 3 developments: CAD PRS testing has become clinically accessible,^{28,29} Lp(a)-lowering therapies are undergoing phase III trials, and inflammation has emerged as a validated therapeutic target for CAD.^{7,8} Together, these developments make biological risk profiling not merely theoretical but both immediately feasible and important for contemporary cardiovascular prevention.

In this study, we evaluated the predictive utility of a single measurement of 4 biomarkers, CAD PRS, LDL-C, hsCRP, and Lp(a), to estimate the risk of CAD by age and sex among individuals aged 40 to 69 years in the UK Biobank (UKB) over a median follow-up of

12 years. By integrating genomics with lipid and inflammatory pathways, we aim to complement traditional risk scores such as the pooled cohort equations (PCE), enabling earlier and more targeted prevention, particularly for younger adults and women, where current risk models underperform and have reduced discrimination.^{30,31}

METHODS

STUDY POPULATION. This study used data from the UKB, a national study that enrolled participants aged 40 to 69 years between 2006 and 2010, collecting detailed phenotypic, genomic, and blood sample data from >500,000 individuals. Subjects who were free from CAD at baseline, not on statin therapy, and had available biomarker and genomic information at baseline were included in the analysis. We excluded subjects with established CAD or cardiovascular events, lacking quality-controlled genotyping, missing baseline biomarker information, and on statin therapy at baseline (**Central Illustration, Figure 1**). Analyses of the UKB were performed under application 7089, in accordance with an Institutional Review Board-approved protocol at Massachusetts General Hospital.

EXPOSURES. This study investigated the association between CAD PRS, hsCRP, Lp(a), and LDL-C and incident CAD across age and sex groups (**Central Illustration**). CAD PRS was calculated using the established multiple-ancestry, genome-wide association studies for CAD by Patel et al,¹⁸ which aggregates the cumulative impact of ~1.2 million genetic variants into a single quantitative predictor.

Baseline Lp(a), LDL-C, and hsCRP levels were measured using validated assays previously used for UKB samples.³² Serum Lp(a) and hsCRP levels were calculated from subjects during enrollment through an immunoturbidimetric assay by Randox Laboratories (using a Beckman Coulter AU5800 analyzer). LDL-C levels were calculated from patient serum samples using the enzymatic selective protection method with the Beckman Coulter AU5800 analyzer.

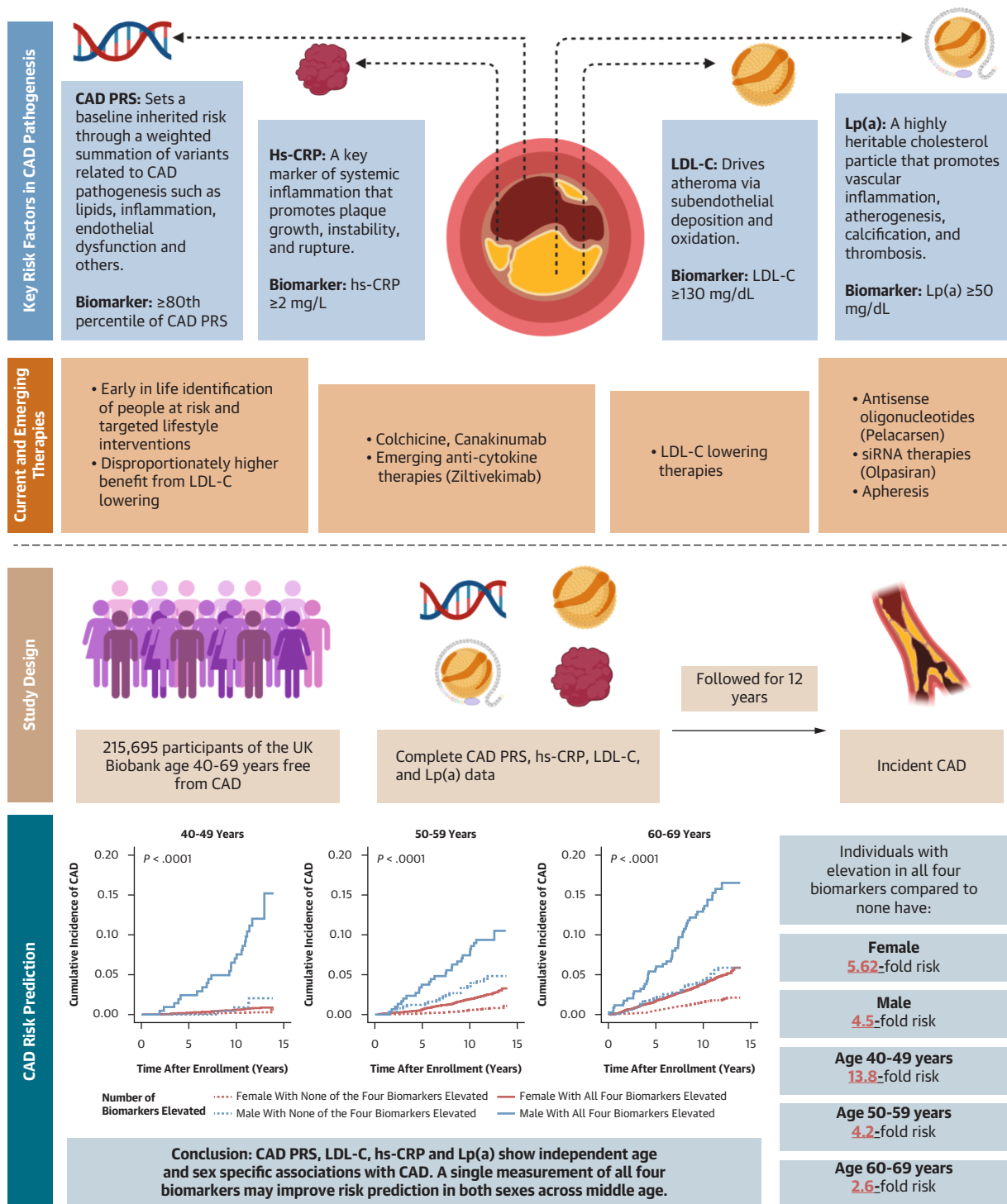
The PCE score was calculated using the 10-year baseline survival functions with age- and race-specific baselines. The original beta coefficients for each risk factor were applied exactly as defined in the derivation cohorts. Variables used in the PCE were age, sex, race, total cholesterol, high-density-lipoprotein cholesterol, systolic blood pressure, antihypertensive treatment status, smoking status, and diabetes status, all defined according to the original PCE specification.³³ A numerical value of the

10-year absolute risk of atherosclerotic cardiovascular disease was used for analysis.

ENDPOINTS AND FOLLOW-UP. The primary endpoint was incident CAD, which was defined as a composite outcome of coronary death, coronary revascularization, and myocardial infarction obtained from self-report, hospital admissions, and diagnosis codes. Reported CAD was obtained from information on hospital diagnosis or death records in the biobank. Self-reported diagnosis was used solely to identify prevalent CAD before baseline enrollment for exclusion purposes.

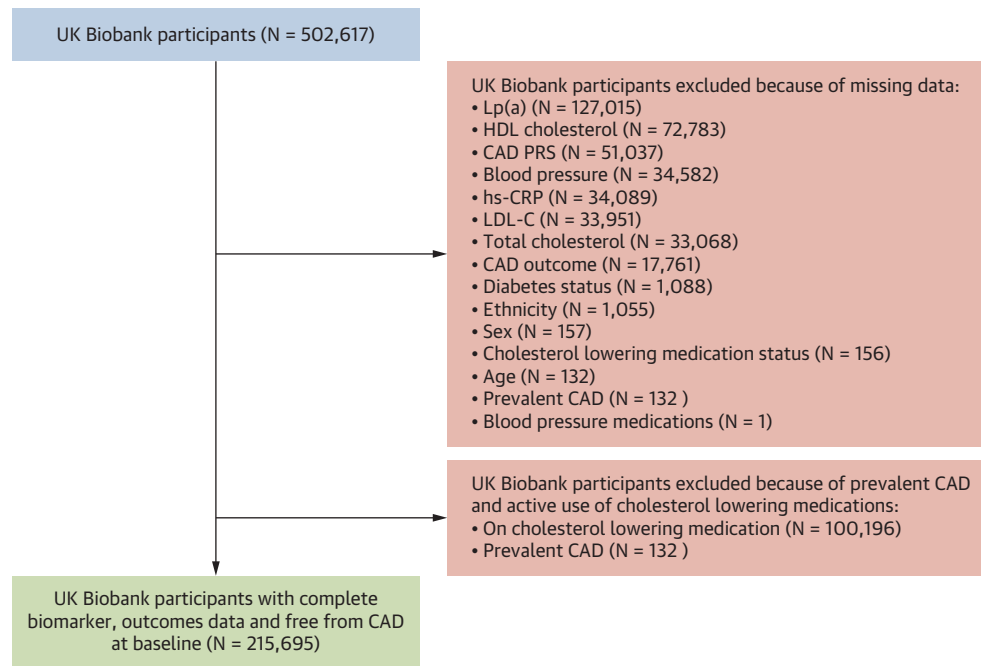
STATISTICAL METHODS. Association of biomarkers with CAD. Continuous measures were evaluated for the Gaussian assumption. Normally distributed data are presented as mean $\pm$ SD and non-normally distributed data as median (IQR). Counts are presented as absolute numbers and proportions (percentages) of the total. Baseline characteristics, including demographic, clinical, and biomarker data, are presented for the entire cohort and stratified by sex. Spearman correlation coefficients for the entire cohort were calculated to assess the correlation between the 4 biomarkers and continuous levels of CAD PRS, LDL-C, hsCRP, and Lp(a).

For our primary analysis, we first modeled biomarkers on a continuous scale to establish their underlying biological relationships with CAD and to assess independence and additivity across domains. All continuous variables were standardized, and HRs for each continuous covariate were expressed per SD increase. A cox proportional hazards model was fitted to the full cohort to assess the association of biomarkers with CAD and test for interactions with age and sex. The model was adjusted for age, sex, all biomarkers, first 10 principal components, diabetes, systolic blood pressure, blood pressure medication, smoking, body mass index, and Townsend deprivation index. We further stratified by age groups (40-49, 50-59, and 60-69 years) and sex to allow age- and sex-specific associations. Restricted cubic spline models were used to examine potential nonlinearity. To translate findings from continuous biological relationships into clinically intuitive measures of association, we evaluated the association of each biomarker with CAD using clinically recognized thresholds. The clinical threshold of each biomarker was defined as a binary variable where an elevated biomarker was defined as CAD PRS $\geq$ 80th percentile of the population distribution, LDL-C $\geq$ 130 mg/dL, hsCRP $\geq$ 2 mg/L, and Lp(a) $\geq$ 125 nmol/L, and comparisons were made to values below these thresholds. The cutoff value for CAD PRS, corresponding to

CENTRAL ILLUSTRATION Integrated Biological Framework Using 4 Biomarkers to Predict Coronary Artery Disease Risk

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Coronary artery disease polygenic risk score (CAD PRS), low-density-lipoprotein cholesterol (LDL-C), high-sensitivity C-reactive protein (hsCRP), and lipoprotein(a) (Lp(a)) are blood biomarkers that independently predict coronary artery disease (CAD). Among 215,695 UK Biobank participants without CAD, those with all 4 biomarkers (CAD PRS, LDL-C, hsCRP, and Lp(a)) elevated had a 5.63-fold (women) and 4.52-fold (men) increase in risk of incident CAD compared with those with none elevated. By age, those with all 4 biomarkers elevated had a 13.79-fold (40-49 years), 4.21-fold (50-59 years), and 2.59-fold (60-69 years) increase in risk of incident CAD compared with those with none elevated. Elevated biomarkers were defined as CAD PRS ≥ 80 th percentile of the population distribution, LDL-C ≥ 130 mg/dL, hsCRP ≥ 2 mg/L, and Lp(a) ≥ 125 nmol/L, and comparisons were made to values below these thresholds. siRNA = small interfering RNA.

FIGURE 1 Study Design Flow Chart

Among 502,617 UK Biobank participants, 282,922 participants were excluded because they had missing data, were on lipid-lowering medications, or had CAD at baseline. After exclusions, 215,695 participants were included in the analysis. CAD = coronary artery disease; CAD PRS = coronary artery disease polygenic risk score; HDL = high-density lipoprotein; hsCRP = high-sensitivity C-reactive protein; LDL-C = low-density-lipoprotein cholesterol; Lp(a) = lipoprotein(a).

the top quintile, was chosen because individuals with these values have a risk equivalent to that of carriers of hypercholesterolemia variants, diabetes, and severe hypercholesterolemia, all of which are risk factors that justify initiating statin treatment under current primary prevention guidelines.^{18,34} HRs were calculated to estimate the risk associated with each elevated biomarker compared with the reference group (not elevated). To account for multiple testing, the significant *P* value was defined as *P* < 0.0125 after Bonferroni multiple testing correction.

For our secondary analysis, biomarker levels in the study population were categorized into 5 quintiles similar to prior studies.^{2,3} Biomarker values were divided into 5 quintiles based on their population distribution (full cohort, age-specific, and sex-specific quintiles). We constructed a Cox proportional hazards model with individual biomarkers and combined biomarkers, where the HR of incident CAD was calculated comparing quintiles 2, 3, 4, and 5 with quintile 1 (reference). HRs per quintile increase in biomarker levels were also calculated. To calculate HRs per quintile increase, biomarker values were

divided into 5 quintiles as described above. An ordinal variable coded 1 through 5 represented quintile groups and was entered into a Cox proportional hazards model as a continuous term and reported per quintile HR reflected the change in CAD risk for each quintile increase in the biomarker level. All survival analyses were stratified by age and sex, using age- and sex-specific biomarker quintiles to assess potential differences.

Finally, we conducted 5 sets of sensitivity analyses. First, we conducted an analysis using Fine-Gray competing risks for subdistribution models to account for competing risks, such as death. Second, we conducted an analysis that included participants who were taking statins at baseline. Third, we conducted an analysis with apolipoprotein B instead of LDL-C. Fourth, we conducted an analysis excluding individuals with an hsCRP > 10 mg/L who likely have acute inflammation. Fifth, we conducted an analysis using a higher Lp(a) threshold of 150 nmol/L.

Combined impact of biomarkers on CAD. A cumulative incidence function curve was studied using the number of elevated biomarkers at the defined

clinical thresholds to assess the combined effect of multiple biomarkers. This analysis was used to translate the continuous biological relationships into clinically intuitive measures of cumulative biological burden. The absolute risk of developing CAD by age and sex was calculated based on the number of modifiable elevated biomarkers (hsCRP, Lp(a), and LDL-C), separately for low and high PRS categories. Absolute risk was calculated using the predict formula for a Cox model with the number of elevated biomarkers as covariates. A sensitivity analysis was conducted to study cumulative incidence function curves for the number of elevated biomarkers using levels in the fifth quintile rather than the clinical threshold, stratified by age and sex groups. Incidence of CAD was calculated when all 4 biomarkers were elevated compared with when none of the biomarkers were elevated and stratified by age and sex groups. Additionally, HRs were estimated in a Cox proportional hazards model according to the number of elevated biomarkers, ranging from 0 (reference) to 4, both in the entire cohort and stratified by age and sex.

Evaluating the performance of the combined 4-biomarker model. Next, we evaluated the performance of the combined 4-biomarker model (CAD PRS, hsCRP, Lp(a), and LDL-C) against the base model (age and sex) across 5-year age bins in both sexes. As a supplement, we also evaluated the C-statistics for the base model (age + sex), base model plus individual biomarkers separately, base model plus the 4 biomarkers, and base model plus all biomarkers (excluding LDL-C, because it is part of the PCE) plus PCE. We also evaluated the augmentation in C-statistic in the full cohort when CAD PRS was added to models with age, sex, and each biomarker individually. All models were bootstrapped 100 times, and their optimism-corrected C-statistics, along with their respective 95% CIs, were reported. Calibration plots were plotted for the Cox model, which includes the combined 4 biomarkers (CAD PRS, hsCRP, Lp(a), and LDL-C) and the first 10 principal components. Calibration plots were plotted across age and sex groups.

Next, we evaluated the net reclassification index (NRI) of a model incorporating age, sex, and the 4 biomarkers (CAD PRS, Lp(a), LDL-C, and hsCRP) in comparison with a model consisting of PCE alone. The NRI was assessed over a 10-year follow-up, using a risk threshold of 7.5% for categorical reclassification and 0 for continuous reclassification. The CI for NRI was calculated through bootstrapping 100 times. Additionally, we calculated the NRI for models consisting of PCE combined with each biomarker

separately (excluding LDL-C, because it is already incorporated into the PCE), compared with PCE alone, to assess the incremental reclassification improvement for each biomarker when added to the PCE. We also calculated the NRI of a model consisting of PCE with all 4 biomarkers (excluding LDL-C, because it is already included in PCE) compared with PCE alone.

Population attributable risk analyses. To assess the relative importance of CAD PRS compared with other biomarkers, population attributable risk (PAR) was calculated for each biomarker across age and sex. PAR is computed using the function $1 - [1 - L(t)] / [1 - L(t)]$, where $L(t)$ is the counterfactual logistic model if the risk factor is absent and $L(t)$ is the logistic function if the risk factor is present. PAR is calculated using multivariable logistic regression models, where risk factors are treated as polychotomous exposures. The lowest quintile (quintile 1) is considered unexposed, and quintiles 2 to 5 are considered exposed, as previously described.²⁰ All multivariable logistic models were adjusted for age, sex, all biomarkers, first 10 principal components of genetic ancestry, diabetes, systolic blood pressure, blood pressure medication, smoking, race, total cholesterol, high-density-lipoprotein cholesterol, body mass index, alcohol use, sleep duration, estimated glomerular filtration fraction, and Townsend deprivation index. Estimated glomerular filtration rate was calculated using the Chronic Kidney Disease Epidemiology Collaboration Equation.³⁵ The R package *attribrisk*, version 0.1.4 (R Foundation for Statistical Computing) was used for the analysis.

Given the conceptual nature of PAR, all estimates are presented as associational and interpreted as “potentially attributable,” assuming causality of biomarkers to CAD. For genetic risk, PAR reflects the burden of lifelong inherited susceptibility to CAD. In contrast, PAR estimates for LDL-C, hsCRP, and Lp(a) represent domains that are increasingly modifiable by targeted intervention.

Modeling treatment eligibility shifts. To model treatment eligibility shift using the combined 4-biomarker model compared with PCE, we identified adults meeting criteria for stage 1 hypertension and/or borderline LDL-C. Stage 1 hypertension is defined as systolic blood pressure between 130 and 139 mm Hg and/or diastolic pressure between 80 and 89 mm Hg. Borderline LDL-C is defined as having a LDL-C level of between 110 and 129 mg/dL. Within this group, we applied the combined 4-biomarker model (CAD PRS, hsCRP, LDL-C, and Lp(a)) and compared its NRI with PCE alone. For each

participant, we estimated the 10-year CAD risk using both approaches and evaluated changes in risk categorization at the clinical threshold of 7.5% for statin treatment. Reclassification performance was assessed using continuous and binary NRIs. To quantify the clinical impact, we applied identical statin-effect assumptions across models, using a relative risk reduction of 0.25,³⁶ and calculated the absolute risk, absolute risk reduction, and the projected number of CAD events prevented per 100,000 adults screened. Treatment-eligibility shifts reflected the proportion of individuals who newly exceeded the 7.5% threshold under the combined 4-biomarker model; events prevented were calculated by multiplying the absolute risk, relative risk reduction of statin (0.25), and 100,000. This analysis estimates how biological profiling with biomarkers may influence preventive therapy decisions by reclassifying individuals toward a higher risk score, thereby increasing their eligibility for preventive therapy and modeling population effects. This analysis was intended to illustrate how biological risk profiling could reorient preventive therapy decisions and project event reduction at a population level. All statistical analyses were performed using R Studio version 2024.

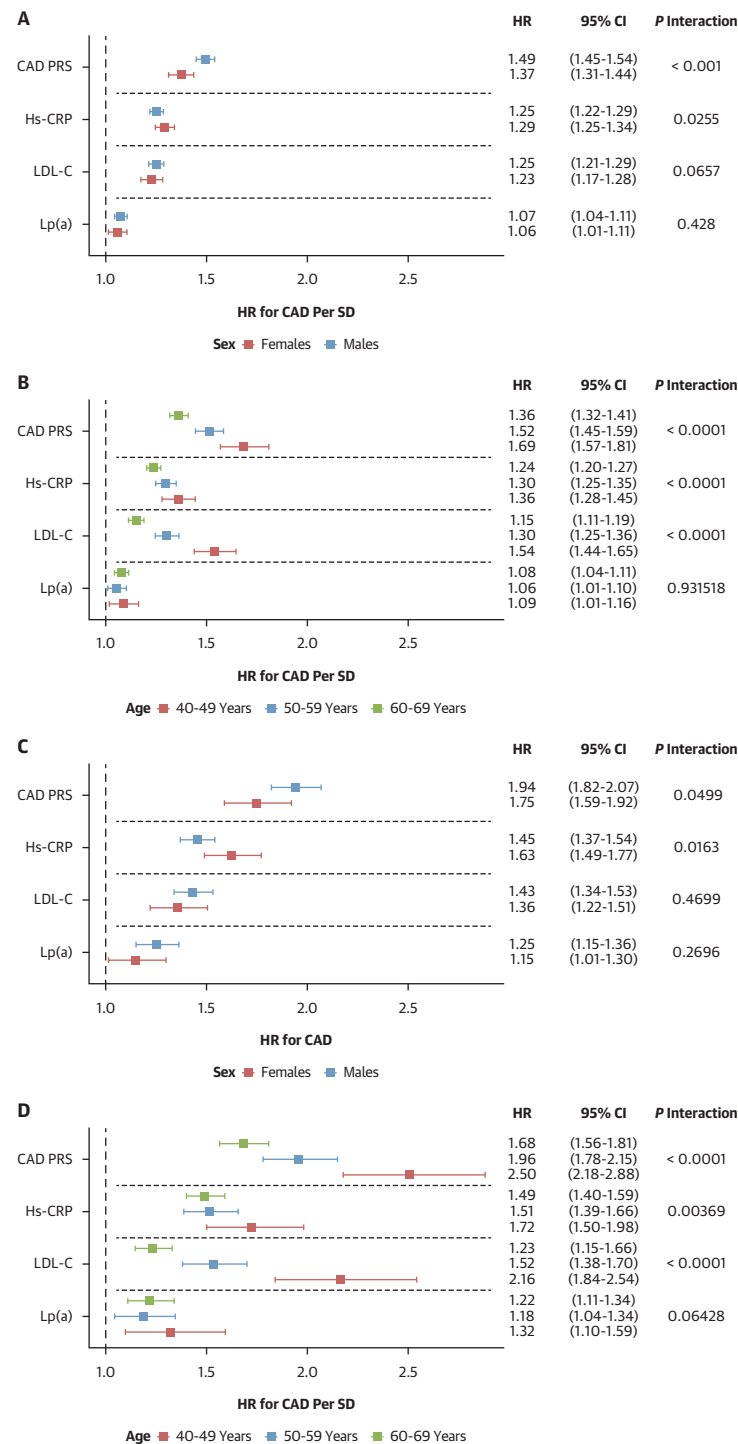
RESULTS

BASELINE CHARACTERISTICS OF THE STUDY POPULATION. Among 215,695 participants (mean age: 55.9 years; 56.0% women; 88.1% White), 6,470 (3.00%) developed CAD over a median follow-up of 12.0 years (IQR: 11.3-12.7 years) (Supplemental Table 1). The mean LDL-C level was 142.1 ± 28.7 mg/dL. The median Lp(a) and hsCRP levels were 20.5 nmol/L (IQR: 9.50-60.5 nmol/L) and 1.29 (IQR: 0.63-2.69 mg/L), respectively. Spearman correlation coefficient analysis showed weak correlation among the 4 biomarkers (Supplemental Figure 1). Sex-specific baseline characteristics showed that of 94,416 men (mean age: 55.9 years; 93.0% White), 4,620 (4.89%) developed CAD over a median follow-up of 12.0 years (IQR: 11.2-12.7 years). Of 121,279 women (mean age: 56.2 years; 93.4% White), 2,111 (1.74%) developed CAD over a median 12.0 years (IQR: 11.4-12.7 years) (Supplemental Table 1). The baseline characteristics, stratified by elevated biomarkers above clinical thresholds and quintiles, are summarized in Table 1 and Supplemental Table 2, respectively.

ASSESSING NONPROPORTIONAL HAZARDS AND NONLINEARITY. We first used continuous modeling to characterize biological relationships between each biomarker and CAD and to demonstrate independence and additivity across domains. Restricted cubic spline models of the association between continuous CAD PRS, log-transformed hsCRP, LDL-C, and log-transformed Lp(a) are shown in Supplemental Figures 2 to 5, demonstrating a linear association with CAD in the full cohort and across ages and sexes, which supports the use of linear terms in the main analysis. Schoenfeld residual analysis revealed that the Cox proportional hazards assumption was violated for biomarkers CAD PRS, hsCRP, and LDL-C but not for Lp(a). However, this violation was resolved in our analysis, which stratifies the Cox model by age groups (Supplemental Figure 6). The log-log survival plots of all biomarkers revealed no violation of the proportional hazards assumption across sex groups; however, it was violated across age groups. Our analysis resolved this issue by stratifying our Cox model by age groups (Supplemental Figure 7).

ASSOCIATION OF CAD PRS, LP(A), hsCRP, AND LDL-C WITH INCIDENT CAD BY AGE AND SEX. As expected, all 4 biomarkers were strongly associated with incident CAD in the full cohort (Supplemental Figures 8 and 9) and across age and sex (Figure 2). On a continuous scale, for every SD increase, the HRs were 1.45 (95% CI: 1.41-1.48) for CAD PRS, 1.21 (95% CI: 1.18-1.24) for LDL-C, 1.05 (95% CI: 1.03-1.08) for Lp(a), and 1.23 (95% CI: 1.20-1.26) for hsCRP (Supplemental Figure 8). In models adjusted for age, sex, and biomarkers, the HRs for incident CAD comparing elevated vs nonelevated biomarkers were 1.79 (95% CI: 1.70-1.89) for CAD PRS, 1.60 (95% CI: 1.48-1.66) for LDL-C, 1.20 (95% CI: 1.12-1.29) for Lp(a), and 1.64 (95% CI: 1.57-1.72) for hsCRP (Supplemental Figure 9).

In sex-stratified analyses, for every SD increase in CAD PRS, men conferred a greater HR of 1.49 (95% CI: 1.45-1.54) compared with women, of 1.37 (95% CI: 1.31-1.44; P -interaction < 0.001) (Figure 2). There were no notable sex differences in the association of LDL-C, Lp(a), and hsCRP with CAD (Figures 2A and 2C). When comparing elevated vs nonelevated biomarkers, the HR for incident CAD was also slightly greater in men (HR: 1.94; 95% CI: 1.82-2.07) than women (HR: 1.75; 95% CI: 1.59-1.92) for CAD PRS (P -interaction < 0.049) (Figure 2C).

FIGURE 2 Association of Biomarkers With Incident CAD Across Age and Sex Groups

HRs of CAD incidence by 4 biomarkers, LDL-C, hsCRP, CAD PRS, and Lp(a), across age and sex groups in a multivariable adjusted model incorporating all the biomarkers. Association analysis was performed using both continuous and categorical variables. Elevated categorical biomarkers are defined as CAD PRS in the top quintile of the population distribution, Lp(a) ≥ 125 nmol/L, LDL-C ≥ 130 mg/dL, and hsCRP ≥ 2 mg/L. The HR was calculated by comparing hazards between elevated and nonelevated biomarkers. Association of the 4 continuous biomarkers with CAD across sex (A) and age (B) groups. Association of the 4 categorical biomarkers with CAD across sex (C) and age (D) groups. The Bonferroni-corrected P value after adjustment for multiple testing was $P < 0.0125$. Abbreviations as in Figure 1.

In age-stratified analyses, the HR for incident CAD was higher in younger ages (40-49 years) compared with older ages (60-69 years) for CAD PRS, LDL-C, and hsCRP but not for Lp(a) (**Figures 2B and 2D**). For every SD increase, the HRs for CAD PRS, LDL-C, and hsCRP, but not Lp(a), were highest at younger age groups (40-49 years) compared with age groups 50 to 59 years and 60 to 69 years (P -interaction < 0.0001) (**Figure 2B**). Comparing the elevated vs nonelevated biomarkers, the HRs for CAD PRS, LDL-C, and hsCRP, but not Lp(a), were highest at younger age groups (40-49 years) compared with the age groups 50 to 59 years and 60 to 69 years (P -interaction < 0.0001). The HR for CAD PRS was 2.50 (95% CI: 2.18-2.88) for 40 to 49 years and 1.68 (95% CI: 1.56-1.81) for 60 to 69 years. For LDL-C, the HR was 2.16 (95% CI: 1.84-2.54) for 40 to 49 years and 1.12 (95% CI: 1.15-1.33) for 60 to 69 years. For Lp(a), the HR was 1.32 (95% CI: 1.10-1.59) for 40 to 49 years and 1.22 (95% CI: 1.11-1.34) for 60 to 69 years. For hsCRP, the HR was 1.72 (95% CI: 1.50-1.98) for 40 to 49 years and 1.49 (95% CI: 1.40-1.59) for 60 to 69 years (**Figure 2D**). In a secondary analysis using quintiles for each biomarker, the results were comparable, and all 4 biomarkers were associated with incident CAD in the full cohort and across age and sex groups (**Supplemental Table 3**).

We conducted 5 sensitivity analyses to ensure the robustness of our findings, all of which demonstrated consistent findings. First, we used Fine-Gray modeling to adjust for death as a competing risk (**Supplemental Figure 10**). Second, we conducted an analysis that included participants on statin at baseline (**Supplemental Figure 11**). Third, we conducted an analysis replacing LDL-C with apolipoprotein B (**Supplemental Figures 12 and 13**). Fourth, we conducted an analysis excluding individuals with likely acute inflammation, specifically those with hsCRP > 10 mg/L (**Supplemental Figure 14**). Fifth, we performed an analysis using a different Lp(a) threshold of 150 nmol/L (**Supplemental Figure 15**).

RELATIVE RISK OF CAD BY NUMBER OF BIOMARKERS ELEVATED ACROSS AGE AND SEX.

The number of elevated biomarkers was strongly associated with incident CAD, with an HR of 4.65 (95% CI: 3.90-5.54) when all biomarkers were elevated compared with when no biomarkers were elevated (**Figure 3A**, **Table 2**, **Supplemental Figure 16A**). Across all age and sex groups, the highest CAD incidence and HR for CAD were observed when all 4 biomarkers were elevated (**Figure 3**, **Central Illustration**).

In sex-stratified analysis, no difference was notable between sex groups, with both men and

women demonstrating higher HRs when all biomarkers were elevated compared with when no biomarkers were elevated, with an HR of 4.52 (95% CI: 3.63-5.63) for men and 5.63 (95% CI: 4.19-7.58) for women (**Table 2**, **Supplemental Figure 16**). Men exhibited a higher incidence of CAD than women, emphasizing sex-related differences in absolute risk of CAD (**Figure 3**, **Central Illustration**).

In age-stratified analysis, individuals in the 40- to 49-year age group with all 4 biomarkers elevated showed a substantially higher cumulative incidence of CAD compared with those with no biomarkers elevated (**Figure 3**, **Central Illustration**), with an HR of 13.79 (95% CI: 8.79-21.14) (**Table 2**, **Supplemental Figure 16**). This pattern was consistent but less pronounced in the 50- to 59-year (HR: 4.21; 95% CI: 3.08-5.75) and 60- to 69-year (HR: 2.59; 95% CI: 2.02-3.33) age groups (**Table 2**, **Figure 3**, **Supplemental Figure 16**, **Central Illustration**).

A secondary analysis counting the number of biomarkers in the top quintile yielded similar results, where the cumulative incidence of and HRs for CAD were highest when all biomarkers were elevated across age and sex groups (**Supplemental Figures 17-19**, **Supplemental Table 4**). To illustrate the clinical utility of evaluating the number of elevated biomarkers at an individual level, we show a random sample of 10 participants with their biomarker levels, highlighting risk driven by variable biological profiles (**Supplemental Figure 20**).

ABSOLUTE RISK OF CAD BY NUMBER OF BIOMARKERS ELEVATED ACROSS AGE AND SEX.

The absolute risk of CAD increased with the increasing number of elevated biomarkers across all age and sex groups (**Figure 4**). To understand the relationship between the 3 modifiable biomarkers (LDL-C, Lp(a), and hsCRP) and the nonmodifiable biomarker, CAD PRS, we evaluated the absolute risk of CAD by the number of elevated modifiable biomarkers (0-3) by high vs nonhigh CAD PRS (**Figure 4**).

In age- and sex-stratified analyses, as expected, men and older people had a higher absolute risk of CAD. However, for each age and sex group, the absolute risk of CAD increased with each additional elevated modifiable biomarker within strata of high and low PRS. The stratification from low PRS with no elevated modifiable biomarkers to high PRS with 3 elevated modifiable biomarkers ranged from 0.48% to 6.40% for age group 40 to 49 years, 1.49% to 6.17% for age group 50 to 59 years, 3.36% to 8.47% for age group 60 to 69 years, 2.67% to 11.5% for men, and 0.80% and 4.40% for women (**Figure 4**). A secondary analysis counting the

TABLE 1 Baseline Characteristics of the Study Population

	Categories		
	Elevated Biomarkers	Nonelevated Biomarkers	P Value
CAD PRS			
Percentile	≥80th	<80th	
Number of participants	42,125	173,570	
Age, y	55.40 ± 8.07	56.06 ± 8.05	<0.001
Female	23,877 (56.7)	97,402 (56.1)	0.036
Incident CAD	2,064 (5.00)	4,667 (3.0)	<0.001
Diabetes	1,049 (2.49)	3,648 (2.10)	0.036
Current smoking	4,690 (11.1)	17,967 (10.4)	<0.001
Total cholesterol, mg/dL	229.19 ± 37.13	225.03 ± 37.01	<0.001
Systolic blood pressure, mm Hg	140.46 ± 20.12	139.44 ± 20.09	<0.001
Diastolic blood pressure, mm Hg	82.59 ± 10.59	82.01 ± 10.61	<0.001
Antihypertension medication	3,777 (8.97)	14,033 (8.08)	<0.001
PCE category			0.020
Low	27,191 (64.5)	110,780 (63.8)	
Intermediate	12,488 (29.6)	52,426 (30.2)	
High	2,446 (5.8)	10,364 (6.0)	
hsCRP			
Plasma levels, mg/dL	≥2	<2	
Number of participants	74,143	141,552	
Age, y	57.00 ± 8.00	55.4 ± 8.04	<0.001
Female	43,744 (59.0)	76,438 (54.0)	<0.001
Incident CAD	2,965 (4.0)	4,246 (3.0)	<0.001
Diabetes	1,482 (2.0)	1,415 (1.0)	<0.001
Current smoking	11,862 (16.0)	16,986 (12.0)	<0.001
Total cholesterol, mg/dL	227.7 ± 37.7	224.9 ± 36.7	<0.001
Systolic blood pressure, mm Hg	143.4 ± 20.3	137.7 ± 19.7	<0.001
Diastolic blood pressure, mm Hg	84.1 ± 10.51	81.1 ± 10.51	<0.001
Antihypertension medication	8,897 (12.0)	8,493 (6.00)	<0.001
PCE category			<0.001
Low	42,336 (57.1)	95,635 (67.6)	
Intermediate	25,468 (34.3)	39,446 (27.9)	
High	6,339 (8.5)	6,471 (4.6)	

Continued on the next page

number of modifiable biomarkers in the top quintile yielded similar results, where the absolute risk of CAD increased with the number of biomarkers in the top quintile across all age and sex groups (Supplemental Figure 21).

PREDICTIVE UTILITY OF CAD PRS, LP(A), hsCRP, AND LDL-C IN PREDICTING CAD ACROSS AGE AND SEX. Each biomarker contributed to an increase in the C-statistic compared with a base model consisting of age and sex. The combined 4-biomarker model integrating continuous CAD PRS, Lp(a), hsCRP, and LDL-C achieved the highest C-statistic (Supplemental Figure 22A). Adding CAD PRS to models, including each modifiable biomarker consistently improved discrimination (Supplemental Figure 22B). A model that included age, sex, and the 4 biomarkers had a

higher C-statistic (0.753) compared with a model with the PCE (0.740); however, a combined model that included both PCE and biomarkers (excluding LDL-C, because it is included in PCE) had the highest C-statistic (0.765) (Supplemental Figure 22A). The C-statistic values of the combined 4-biomarker model compared with a base model (age and sex) were highest among individuals aged 40 to 45 years and declined with increasing age in both sexes (Figures 5A and 5B). The combined 4-biomarker model was well calibrated across age and sex groups (Supplemental Figure 23).

We also compared the net reclassification of a combined 4-biomarker model, which includes age, sex, CAD PRS, Lp(a), and hsCRP, with that of the PCE alone. The NRI was 32.0% (95% CI: 28.8-34.7) and a binary NRI was 4.42% (95% CI: 3.07-5.72) at the threshold of 10-year atherosclerotic cardiovascular disease risk of 7.5%, underscoring improved discrimination of cases for noncases in the combined 4-biomarker model compared with the PCE model (Table 3). A separate analysis, in which PCE was combined with all biomarkers (excluding LDL-C, because it is already included in PCE) compared with PCE alone, yielded a binary NRI of 5.07% (95% CI: 4.11-5.88) (Supplemental Table 5). A separate NRI analysis where PCE was combined with each biomarker (CAD PRS, Lp(a), or hsCRP) compared with PCE alone yielded a binary NRI of 5.07% (95% CI: 4.11- 5.88), 1.00% (95% CI: 0.47- 1.59), and 1.27% (95% CI: 0.79- 1.70), respectively (Supplemental Tables 6-8).

PAR AND EXPLAINED RELATIVE RISK. In PAR analyses, CAD was primarily attributable to genetics (CAD PRS PAR: 38.7%) in a model adjusted for age, sex, all biomarkers, genetic ancestry, diabetes, systolic blood pressure, blood pressure medication, smoking, race, total cholesterol, high-density-lipoprotein cholesterol, body mass index, alcohol use, sleep duration, estimated glomerular filtration rate, and Townsend deprivation index (Figure 6). These were followed by other risk factors, including hsCRP (26.7%), LDL-C (20.9%), and Lp(a) (9.21%).

However, notable age and sex differences were observed in PAR values across individual biomarkers (Figure 6). Across nearly all subgroups, CAD PRS consistently exhibited the highest PAR in men (42.2%) compared with women (31.9%) and particularly in younger age categories, ages 40 to 49 years (48.2%), which declined with increasing age (50-59 years: 43.0%; 60-69 years: 33.3%). hsCRP showed a considerable attributable risk after CAD PRS in both sexes (men: 26.3%; women: 32.9%), with a higher PAR in

younger individuals aged 40 to 49 years (45.0%) compared with older individuals (50-59 years: 23.5%; 60-69 years: 20.17%). LDL-C also emerged as an important contributor to PAR, with the highest relative contribution in the 40- to 49-year (44.0%) and 50- to 59-year (10.5%) age groups, and remained moderately elevated in older ages, specifically 60 to 69 years (5.43%). LDL-C contributed a slightly higher PAR in women (22.3%) than in men (19.7%). Lp(a) consistently showed the lowest PAR across all age groups (40-49 years: 18.8%; 50-59 years: 9.49%; 60-69 years: 7.00%) and sex (men: 9.09%; women: 10.2%) groups but was highest at younger ages (Figure 6).

MODELING TREATMENT ELIGIBILITY SHIFTS AMONG ADULTS WITH STAGE 1 HYPERTENSION OR BORDERLINE LDL-C. We identified a subset of 123,326 individuals with stage 1 hypertension or borderline LDL-C and modeled the shift in statin treatment eligibility when using the combined 4-biomarker model compared with the PCE model. In this subset, the continuous NRI was 32.3% (95% CI: 29.5-36.9) and the binary NRI at the 7.5% risk threshold was 3.88% (95% CI: 2.98-4.86) (Supplemental Table 9). Under identical statin treatment assumptions, where the relative risk reduction of statin was 0.25, the combined 4-biomarker model with PCE would prevent an estimated 2,339 CAD events per 100,000 adults screened compared with 1,445 events using PCE alone (Supplemental Table 10). Individuals reclassified above the 7.5% threshold had a higher observed absolute risk (9.36%) than those identified by PCE alone (5.78%) (Supplemental Table 10). The absolute risk of CAD declined progressively with statin treatment across biomarker strata, with the greatest reduction (absolute risk reduction: 2.2%) among those identified as having a risk of $\geq 7.5\%$ by the combined 4-biomarker model (Supplemental Figure 24).

DISCUSSION

In this study of 215,695 middle-aged participants free of CAD, a 4-biomarker-based screening consisting of CAD PRS, hsCRP, Lp(a), and LDL-C predicted incident CAD over a follow-up period of >10 years in both sexes and across the entire age spectrum, starting at 40 years of age. This study reinforces the value of integrating biomarker-based risk profiling and traditional risk scores to improve early identification of individuals at high risk of CAD. Our findings support a reframing of CAD risk from a predominantly short-term, clinically based prediction model to a life-course, biologically driven framework that

TABLE 1 Continued

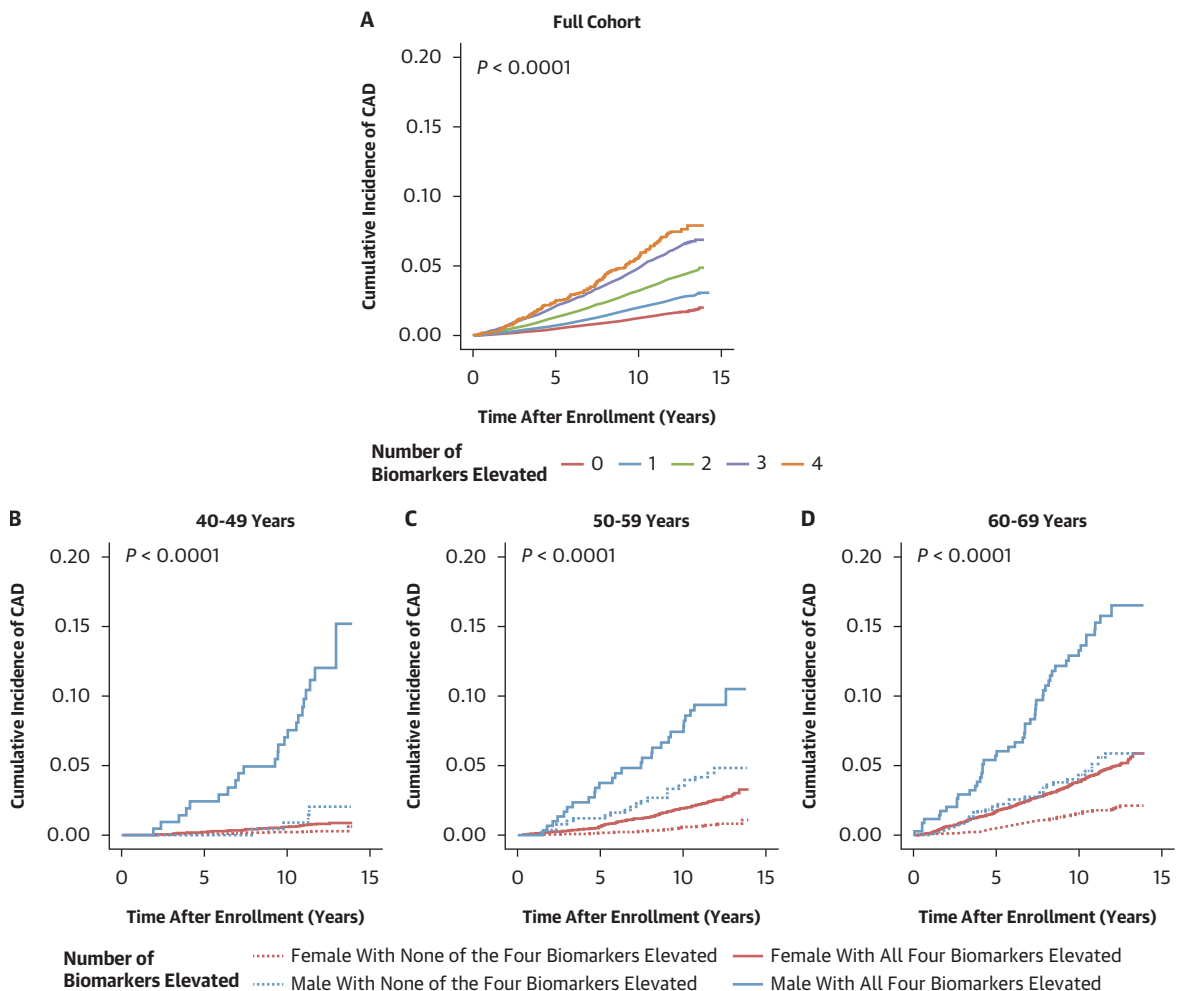
	Categories		
	Elevated Biomarkers	Nonelevated Biomarkers	P Value
Lp(a)			
Plasma levels, nmol/L	≥ 125	< 125	
Number of participants	23,208	192,487	
Age, y	56.0 ± 7.96	55.9 ± 8.07	0.327
Female	13,460 (58.0)	107,792 (56.0)	< 0.001
Incident CAD	928 (4.00)	5,774 (3.00)	< 0.001
Diabetes	232 (1.00)	1,924 (1.00)	0.011
Current smoking	3,249 (14.0)	25,023 (13.0)	0.127
Total cholesterol, mg/dL	233.08 ± 36.0	224.97 ± 37.1	< 0.001
Systolic blood pressure, mm Hg	139.9 ± 19.9	139.6 ± 20.1	0.066
Diastolic blood pressure, mm Hg	82.3 ± 10.54	82.11 ± 10.62	0.011
Antihypertension medication	1,856 (8.00)	15,398 (8.00)	0.334
PCE category			< 0.001
Low	15,046 (64.8)	122,925 (63.9)	
Intermediate	6,937 (29.9)	57,977 (30.1)	
High	1,225 (5.3)	11,585 (6.0)	
LDL-C			
Plasma levels, mg/dL	≥ 130	< 130	
Number of participants	139,613	76,082	
Age, y	56.97 ± 7.72	54.02 ± 8.31	< 0.001
Female	76,787 (55.0)	44,127 (58.0)	< 0.001
Incident CAD	5,584 (4.00)	1,521.64 (2.00)	< 0.001
Diabetes	1,396 (1.00)	1,521.64 (2.00)	< 0.001
Current smoking	18,149 (13.0)	9,890 (13.0)	< 0.676
Total cholesterol, mg/dL	245.73 ± 27.6	189.36 ± 21.1	< 0.001
Systolic blood pressure, mm Hg	141.6 ± 19.9	136.1 ± 20.0	< 0.001
Diastolic blood pressure, mm Hg	83.1 ± 10.5	80.3 ± 10.6	< 0.001
Antihypertension medication	11,169 (8.0)	6,086 (8.0)	0.358
PCE category			< 0.001
Low	81,429 (58.3)	56,542 (74.3)	
Intermediate	48,366 (34.6)	16,548 (21.8)	
High	9,818 (7.00)	2,992 (3.90)	

Values are n (%) or mean $\pm$ SD, unless otherwise indicated. Baseline characteristics of 215,695 UK Biobank participants included in the analysis by hsCRP, Lp(a), and LDL-C clinical thresholds. Elevated biomarkers are defined using their clinical thresholds, where a high CAD PRS was defined as the top quintile of the population distribution. High Lp(a) was defined as ≥ 125 nmol/L, high LDL-C as a level of ≥ 130 mg/dL, hsCRP as ≥ 2 mg/L.

CAD = coronary artery disease; CAD PRS = coronary artery disease polygenic risk score; hsCRP = high-sensitivity C-reactive protein; LDL-C = low-density-lipoprotein cholesterol; Lp(a) = lipoprotein(a); PCE = pooled cohort equations.

integrates inherited (CAD PRS) and modifiable pathways (hsCRP, Lp(a), and LDL-C) across age and sex. This reframing is particularly timely given the availability of clinically available PRS testing, emerging Lp(a)-lowering therapies, and treatable inflammatory risk, which together make biological risk profiling actionable. This study has at least 3 implications.

First, using continuous modeling, we found that each of the 4 biomarkers, CAD PRS, hsCRP, Lp(a), and LDL-C, was independently associated with incident CAD, yet their relative contributions varied slightly

FIGURE 3 Cumulative Incidence of CAD by Number of Elevated Biomarkers Stratified by Age and Sex

Cumulative incidence of CAD by number of elevated biomarkers (0 or 4) above the clinical threshold over a follow-up period of 10 years. Cumulative incidence is shown for the full cohort (A) and stratified by age and sex groups (B-D). Clinical thresholds for biomarker elevation are defined as CAD PRS in the top quintile of the population distribution, Lp(a) ≥ 125 nmol/L, LDL-C ≥ 130 mg/dL, and hsCRP ≥ 2 mg/L. Cumulative incidence for the full cohort was derived from a Cox proportional hazards model adjusted for age, sex, and genetic ancestry. Abbreviations as in [Figure 1](#).

by age and sex. CAD PRS conferred the strongest association with incident CAD compared with other biomarkers in both sexes. Although CAD PRS had a stronger association with incident CAD in men compared with women, this may reflect a limitation of the CAD PRS itself.³⁷ PRS designed to include alleles with sex-specific effects are likely to reduce this limitation. Still, its predictive utility was highest earlier in the life course in both men and women, consistent with prior observations.^{17,27,38,39} Younger individuals have fewer environmental exposures and clinical risk factors than older individuals, resulting

in a higher genetic contribution to determining CAD risk. Although genetic risk stratification includes genetic drivers of cholesterol and inflammation pathways, CAD PRS confers a sizable independent genetic risk that persists even after adjusting for LDL-C, Lp(a), and hsCRP. Although each of the modifiable biomarkers, LDL-C, hsCRP, and Lp(a), indicates measurable and targetable mechanisms with pharmacological therapies, the additive risk captured by CAD PRS points to mechanisms of disease not captured in a single measurement of the 3 modifiable blood biomarkers. These could include the genetic

TABLE 2 CAD HRs by Number of Elevated Biomarkers

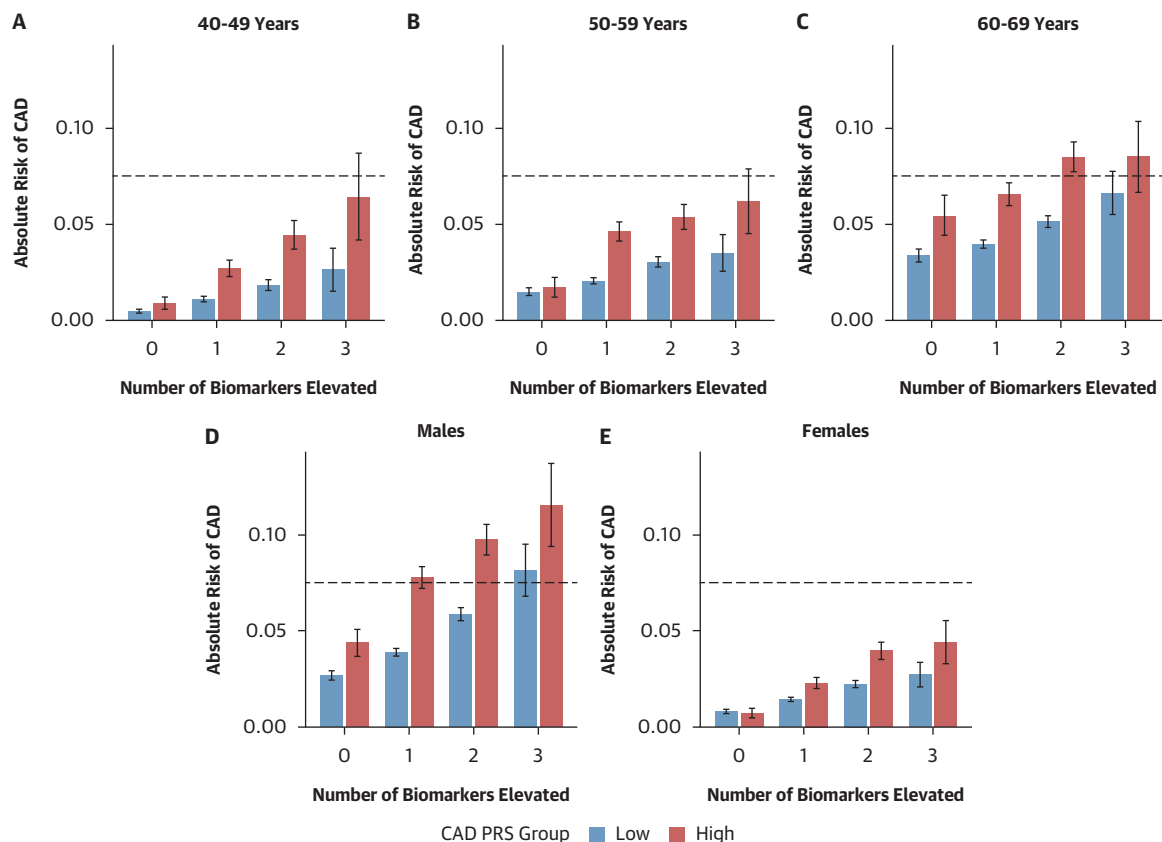
	No. of Elevated Biomarkers				
	0	1	2	3	4
All	1.0 (reference)	1.60 (1.47-1.75)	2.58 (2.37-2.82)	3.89 (3.53-4.29)	4.65 (3.90-5.54)
Women	1.0 (reference)	1.74 (1.48-2.05)	2.86 (2.43-3.35)	4.71 (3.95-5.62)	5.63 (4.19-7.58)
Men	1.0 (reference)	1.47 (1.32-1.63)	2.47 (2.24-2.75)	3.66 (3.25-4.12)	4.52 (3.63-5.63)
Aged 40-49 y	1.0 (reference)	2.30 (1.78-2.97)	4.61 (3.58-5.91)	8.73 (6.63-11.49)	13.79 (8.97-21.14)
Aged 50-59 y	1.0 (reference)	1.35 (1.16-1.58)	2.38 (2.04-2.78)	3.37 (2.82-4.02)	4.21 (3.08-5.75)
Aged 60-69 y	1.0 (reference)	1.21 (1.08-1.36)	1.66 (1.48-1.86)	2.46 (2.16-2.81)	2.59 (2.02-3.33)

Values are HR (95% CI). HR of CAD incidence through the number of elevated biomarkers (≥ 80 th percentile for CAD PRS, ≥ 2 mg/L for hsCRP, ≥ 125 nmol/L for Lp(a), and >130 mg/dL for LDL-C) across age and sex groups.

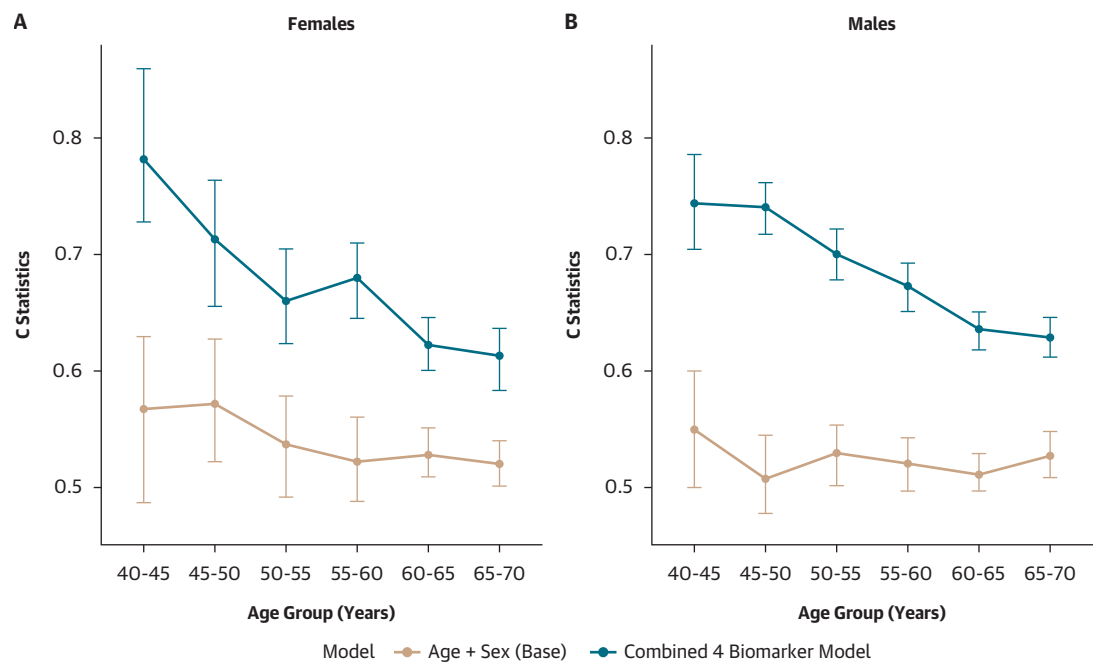
Abbreviations as in Table 1.

susceptibility to lifetime exposure to the biomarker rather than the 1-time measurement, as demonstrated previously with LDL-C⁴⁰⁻⁴³ gene-environment interactions, or additional mechanisms

of disease that are not yet fully understood. Of more than 250 genetic loci identified in the most recent genomewide association study of CAD, 20% were functionally linked to lipid and cholesterol

FIGURE 4 Absolute Risk of CAD Associated With the Number of Elevated Modifiable Biomarkers by CAD PRS

Absolute risk of CAD by number of biomarkers elevated (≥ 2 mg/L for hsCRP, ≥ 125 nmol/L for Lp(a), and ≥ 130 mg/dL for LDL-C) stratified by high (red, ≥ 80 th percentile) vs low (blue, <80 th percentile) CAD PRS and for different age and sex groups (A-E). Abbreviations as in Figure 1.

FIGURE 5 Discrimination of a Combined 4-Biomarker-Based Model by Age and Sex

(A) Discrimination of a combined 4-biomarker model consisting of age, sex, and all 4 biomarkers (coronary artery disease polygenic risk score, high sensitivity-C-reactive protein, lipoprotein(a), and low-density-lipoprotein cholesterol) compared with a base model consisting of age and sex only is shown in women within strata. (B) Similar comparison is shown for men across age strata. C-statistic calculations were subject to bootstrapping 100 times, and the optimism-corrected C-statistics were plotted with their associated 95% CIs.

metabolism, 15% to inflammation and immune pathways, and the remainder were distributed among vascular development, cell cycle signaling, and other uncharacterized processes.⁴⁴ Similarly, there was an independent association of hsCRP even after controlling for LDL-C, Lp(a), and CAD PRS,

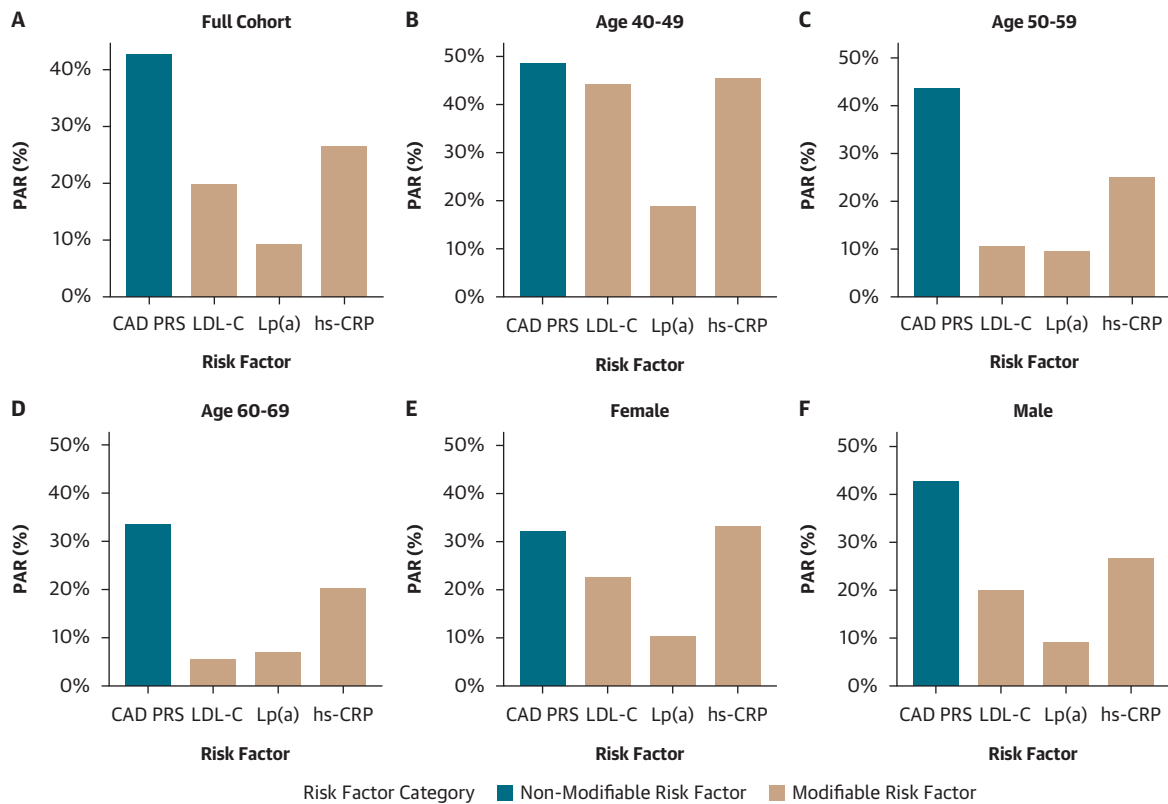
which aligns with current studies on residual inflammatory risk in CAD.²⁰ Lp(a) was associated with CAD in both sexes and across age groups similarly, highlighting that a single measurement of Lp(a) at any age is a robust predictor of CAD. Consistent with these findings, we observed only weak pairwise

TABLE 3 Net Reclassification of the Combined 4-Biomarker Model Compared With PCE

Age, sex, CAD PRS, Lp(a), LDL-C, and hsCRP				
PCE				Total No. of Participants
		<7.5%	≥7.5%	
CAD events	<7.5%	3,870	632	4,502
	≥7.5%	353	402	755
	Total no. of participants	4,223	1,034	5,257
CAD nonevents	<7.5%	189,603	6,171	195,774
	≥7.5%	4,203	2,432	6,635
	Total no. of participants	193,806	8,603	202,409

A reclassification table of the 10-year risk of CAD using age, sex, CAD PRS, Lp(a), and hsCRP compared with the PCE alone is demonstrated by the number of individuals in each risk category. Dark blue represents the number of participants correctly reclassified with the combined model. Light blue represents the number of participants incorrectly reclassified by the combined model. The continuous net reclassification index was 32.0% (95% CI: 28.8-34.7). The risk threshold for both models is 7.5%. The binary NRI at this risk threshold was 4.42% (95% CI: 3.07-5.72).

Abbreviations as in Table 1.

FIGURE 6 PAR of CAD for Each Biomarker by Age and Sex

The PAR of CAD, as determined by different biomarkers, is represented in the full cohort (A) and across age (B-D) and sex groups (E and F). PAR was calculated using multivariable logistic regression models where risk factors were treated as polychotomous exposures, where the lowest quintile (quintile 1 is considered unexposed) and quintiles 2 to 5 were considered exposed. All multivariable logistic models were adjusted for all biomarkers, the first 10 principal components of genetic ancestry, diabetes, systolic blood pressure, blood pressure medication, smoking, race, total cholesterol, high-density lipoprotein cholesterol, body mass index, alcohol consumption, sleep duration, estimated glomerular filtration fraction, and Townsend deprivation index. The multivariable logistic model for the full cohort was further adjusted for age and sex. PAR = population attributable risk; other abbreviations as in [Figure 1](#).

correlation among CAD PRS, LDL-C, hsCRP, and Lp(a), and each biomarker remained independently associated with CAD after mutual adjustment. Together, these indicate that genetic, lipid, and inflammatory risk capture largely nonoverlapping biology. These observations recast CAD risk as a joint expression of inherited susceptibility and modifiable biological pathways rather than as a snapshot of traditional risk factors alone. By quantifying these domains concurrently, our study moves beyond describing individual biomarkers to proposing a unified biological architecture of CAD risk.

Second, the combined 4-biomarker model exerts disproportionately greater impact in younger individuals. Our results demonstrated that the number of elevated biomarkers above clinical thresholds was strongly associated with CAD risk and that a

combined 4-biomarker model augmented risk stratification beyond a clinical risk calculator, particularly in younger individuals. There was a 4.65-fold increased risk of CAD when all 4 biomarkers were elevated compared with when no biomarkers were elevated, and the risk was as high as 13.79-fold among individuals aged 40 to 49 years. A model comprising age, sex, and the 4 biomarkers demonstrated higher discrimination compared with a model consisting of PCE in both sexes, resulting in a meaningful reclassification of events. The combined biomarker model also had the highest discrimination in younger age groups, precisely where improved stratification has the highest preventive potential. Consistent with this, in 123,326 individuals with stage 1 hypertension or borderline LDL-C, the combined 4-biomarker model achieved a continuous NRI of 32.3%

at the 7.5% statin-treatment threshold and under identical statin assumptions was projected to prevent 2,339 CAD events per 100,000 adults screened vs 1,445 using PCE alone, with higher observed absolute risk among those reclassified above 7.5% (9.36% vs 5.78%). These observations collectively reinforce the utility of universal biomarker screening for risk stratification, particularly in younger individuals, where clinical risk factors may not yet have emerged. Prior work has recently established this concept for the 3 modifiable biomarkers, hsCRP, Lp(a), and LDL-C, in the Women's Health Study² and EPIC-Norfolk³ study. Here, we extend this framework by including genetic risk and demonstrate robust augmentation in risk stratification across sex and age groups in >200,000 individuals followed for over a decade, thereby integrating inherited and modifiable pathways into a coherent, life-course model. Conceptually, this shifts risk assessment based on clinical scores to a graded model of biological burden in which each additional abnormal pathway (genetic, lipid, or inflammatory) adds a measurable increment in hazard, where the accumulation of inherited and modifiable risk defines lifetime hazard, especially early in the life course.

Third, biomarker-based risk stratification enables identification of different mechanisms of risk that can be targeted early in the life course. A model that included age, sex, and the 4 biomarkers improved discrimination compared with the age-and-sex-only model and the PCE model, with greater discrimination in younger adults, and was well calibrated across age and sex. The 4-biomarker model improved the continuous NRI of 32.0% compared with PCE. Such an approach aligns with the goals of precision medicine by enabling early risk stratification, not only based on conventional scores to determine candidacy for statin therapy, but also on biologically driven pathways such as inflammation and genetically mediated lipid abnormalities. In this view, risk assessment becomes not just an estimate of event probability, but a biological "map" that links specific pathways to specific preventive options over time. Given the availability of current and emerging therapies targeting specific biomarker pathways, a biomarker-based model may enable translation of risk identification into tailored preventive strategies. Our analysis of PAR helps frame how these pathways contribute to disease at the population level. CAD PRS, which shows the highest PAR in all ages and the highest at younger ages, underscores that inherited susceptibility creates a baseline risk for CAD early in life. Individuals with high CAD PRS have been shown

to have disproportionately favorable risk reduction from lifestyle interventions and lipid-lowering.⁴⁵⁻⁴⁷ hsCRP was the second-highest PAR across all ages and sexes, highlighting the substantial contribution of inflammation to CAD. Elevated hsCRP may identify patients who would particularly benefit from anti-inflammatory therapies, as demonstrated in clinical trials to reduce the risk of CAD.⁶⁻⁸ One anti-inflammatory, low-dose colchicine, is currently approved for preventing CAD.^{7,8} Although Lp(a) contributed the smallest PAR, individuals with elevated Lp(a) are at greater risk of CAD and may soon have access to targeted therapies currently in phase III clinical trials.⁴⁸⁻⁵¹ Finally, a biomarker risk stratification strategy could serve as an enrichment strategy to identify individuals with a high probability of having subclinical coronary plaque, which can be detected with coronary computed tomography angiography. Thus, the proposed framework suggests a shift from reacting to events in middle or late life to proactively aligning biological risk profiles with pathway-specific interventions beginning much earlier.

STUDY LIMITATIONS. This study should be interpreted in the context of several limitations. First, the UKB enrolled middle-aged individuals who, on average, are healthier than the UK population.⁵² Individuals who might have experienced CAD events and are at the highest risk may have been under-represented. This may lead to conservative estimates of the true association between biomarkers and disease. Second, the value of biomarker-based risk stratification could not be determined in individuals aged <40 years in this study. Third, the generalizability of our findings could be improved by population diversity. The study cohort was predominantly of European ancestry, which may limit the applicability to individuals of non-European ancestry. Fourth, because we did not have an independent evaluation cohort, our C-index and NRI estimates may be optimistic. Hence, external validation with another cohort would provide a more rigorous assessment of predictive performance.

CONCLUSIONS

This study highlights the potential utility of 4 blood biomarkers, CAD PRS, LDL-C, Lp(a), and hsCRP, for CAD risk prediction. Each biomarker had a varying impact on predicting CAD across different age and sex groups. Together, these 4 markers define an integrated biological framework for coronary risk that unites genetic, lipid, and inflammatory pathways,

showing its greatest incremental value in younger adults, and makes risk intuitive as a cumulative burden that rises as more markers are abnormal. A single measurement of those 4 biomarkers provided a powerful risk stratification tool that outperformed a clinical risk calculator. Incorporating genetics, lipid, and inflammatory markers into routine risk assessment may enable the early identification of high-risk individuals and support targeted primary prevention strategies.

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is a co-founder of Goodpath and Avigena; and serves as a scientific advisor to MyOme, Arboretum Health, HeartFlow, and Aduitum Bio. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: CAD remains the leading cause of death worldwide, and a comprehensive risk assessment with targetable biomarkers needs to be implemented earlier in life. CAD PRS, LDL-C, Lp(a), and hsCRP are blood biomarkers that together define an integrated biological framework to predict CAD. One-time screening of CAD PRS, LDL-C, hsCRP, and Lp(a) among UKB participants, particularly in younger adults, further enhanced the prediction of individuals at high risk for incident CAD, underscoring the need for comprehensive risk assessment beyond traditional risk factors. The identification of high-risk individuals with high cumulative biological burden allows clinicians to guide personalized, pathway-specific preventive management for CAD.

TRANSLATIONAL OUTLOOK: Additional research is needed to assess the clinical benefits of 1-time screening for CAD PRS, LDL-C, hsCRP, and Lp(a) in diverse populations and to test whether targeting these biological domains improves long-term CAD outcomes.

REFERENCES

- Lloyd-Jones DM, Albert MA, Elkind M. The American Heart Association's focus on primordial prevention. *Circulation*. 2021;144(15):e233–e235. <https://doi.org/10.1161/CIRCULATIONAHA.121.057125>
- Ridker PM, Moorthy MV, Cook NR, Rifai N, Lee IM, Buring JE. Inflammation, cholesterol, lipoprotein(a), and 30-year cardiovascular outcomes in women. *N Engl J Med*. 2024;391(22):2087–2097. <https://doi.org/10.1056/NEJMoa2405182>
- Kraaijenhof JM, Nurmohamed NS, Nordestgaard AT, et al. Low-density lipoprotein cholesterol, C-reactive protein, and lipoprotein(a) universal one-time screening in primary prevention: the EPIC-Norfolk study. *Eur Heart J*. 2025;46(39):3875–3884. <https://doi.org/10.1093/eurheartj/ehaf209>
- Ridker PM. How common is residual inflammatory risk? *Circ Res*. 2017;120(4):617–619. <https://doi.org/10.1161/CIRCRESAHA.116.310527>
- Mora S, Ridker PM. Justification for the use of statins in primary prevention: an intervention trial evaluating rosuvastatin (JUPITER)—can C-reactive protein be used to target statin therapy in primary prevention? *Am J Cardiol*. 2006;97(2A):33A–41A. <https://doi.org/10.1016/j.amjcard.2005.11.014>
- Ridker PM, Everett BM, Thuren T, et al. Anti-inflammatory therapy with canakinumab for atherosclerotic disease. *N Engl J Med*. 2017;377(12):1119–1131. <https://doi.org/10.1056/NEJMoa1707914>
- Tardif JC, Kouz S, Waters DD, et al. Efficacy and safety of low-dose colchicine after myocardial infarction. *N Engl J Med*. 2019;381(26):2497–2505. <https://doi.org/10.1056/NEJMoa1912388>
- Nidorf SM, Fiolet ATL, Mosterd A, et al. Colchicine in patients with chronic coronary disease. *N Engl J Med*. 2020;383(19):1838–1847. <https://doi.org/10.1056/NEJMoa2021372>
- Tsimikas S, Willerson JT, Ridker PM. C-reactive protein and other emerging blood biomarkers to optimize risk stratification of vulnerable patients. *J Am Coll Cardiol*. 2006;47(8 Suppl):C19–C31. <https://doi.org/10.1016/j.jacc.2005.10.066>
- May M. Eleven clinical trials that will shape medicine in 2026. *Nat Med*. 2025;31(12):3943–3947. <https://doi.org/10.1038/s41591-025-04083-x>
- Kronenberg F, Mora S, Stroes ESG, et al. Lipoprotein(a) in atherosclerotic cardiovascular disease and aortic stenosis: a European Atherosclerosis Society consensus statement. *Eur Heart J*. 2022;43(39):3925–3946. <https://doi.org/10.1093/eurheartj/ehac361>
- Mach F, Baigent C, Catapano AL, et al. 2019 ESC/EAS guidelines for the management of dyslipidaemias: lipid modification to reduce cardiovascular risk. *Eur Heart J*. 2020;41(1):111–188. <https://doi.org/10.1093/eurheartj/ehz455>
- Ridker PM. Clinician's guide to reducing inflammation to reduce atherothrombotic risk: JACC Review Topic of the Week. *J Am Coll Cardiol*.

- 2018;72(25):3320–3331. <https://doi.org/10.1016/j.jacc.2018.06.082>
14. Liuzzo G, Ridker PM. Universal screening for hsCRP in patients with atherosclerotic disease: a major therapeutic opportunity. *Eur Heart J*. 2024;45(44):4731–4733. <https://doi.org/10.1093/eurheartj/ehae565>
15. Mazhar F, Faucon AL, Fu EL, et al. Systemic inflammation and health outcomes in patients receiving treatment for atherosclerotic cardiovascular disease. *Eur Heart J*. 2024;45(44):4719–4730. <https://doi.org/10.1093/eurheartj/ehae557>
16. Ridker PM, Lei L, Louie MJ, et al. Inflammation and cholesterol as predictors of cardiovascular events among 13 970 contemporary high-risk patients with statin intolerance. *Circulation*. 2024;149(1):28–35. <https://doi.org/10.1161/CIRCULATIONAHA.123.066213>
17. Marston NA, Pirruccello JP, Melloni GEM, et al. Predictive utility of a coronary artery disease polygenic risk score in primary prevention. *JAMA Cardiol*. 2023;8(2):130–137. <https://doi.org/10.1001/jamacardio.2022.4466>
18. Patel AP, Wang M, Ruan Y, et al. A multi-ancestry polygenic risk score improves risk prediction for coronary artery disease. *Nat Med*. 2023;29(7):1793–1803. <https://doi.org/10.1038/s41591-023-02429-x>
19. O'Sullivan JW, Raghavan S, Marquez-Luna C, et al. Polygenic risk scores for cardiovascular disease: a scientific statement from the American Heart Association. *Circulation*. 2022;146(8):e93–e118. <https://doi.org/10.1161/CIR.0000000000001077>
20. Bhattacharya R, Marnell CS, Cho SMJ, et al. Importance of clinical, laboratory, and genetic risk factors for incident CAD. *Circ Genom Precis Med*. 2025;18(4):e004937. <https://doi.org/10.1161/CIRCGEN.124.004937>
21. Inouye M, Abraham G, Nelson CP, et al. Genomic risk prediction of coronary artery disease in 480,000 adults: implications for primary prevention. *J Am Coll Cardiol*. 2018;72(16):1883–1893. <https://doi.org/10.1016/j.jacc.2018.07.079>
22. Abraham G, Havulinna AS, Bhalala OG, et al. Genomic prediction of coronary heart disease. *Eur Heart J*. 2016;37(43):3267–3278. <https://doi.org/10.1093/eurheartj/ehw450>
23. Watkins H, Farrall M. Genetic susceptibility to coronary artery disease: from promise to progress. *Nat Rev Genet*. 2006;7(3):163–173. <https://doi.org/10.1038/nrg1805>
24. Khera AV, Kathiresan S. Genetics of coronary artery disease: discovery, biology and clinical translation. *Nat Rev Genet*. 2017;18(6):331–344. <https://doi.org/10.1038/nrg.2016.160>
25. Riveros-Mckay F, Weale ME, Moore R, et al. Integrated polygenic tool substantially enhances coronary artery disease prediction. *Circ Genom Precis Med*. 2021;14(2):e003304. <https://doi.org/10.1161/CIRCGEN.120.003304>
26. Hindy G, Aragam KG, Ng K, et al. Genome-wide polygenic score, clinical risk factors, and long-term trajectories of coronary artery disease. *Arterioscler Thromb Vasc Biol*. 2020;40(11):2738–2746. <https://doi.org/10.1161/ATVBAHA.120.314856>
27. Urbut SM, Cho SMJ, Paruchuri K, et al. Dynamic importance of genomic and clinical risk for coronary artery disease over the life course. *Circ Genom Precis Med*. 2025;18(1):e004681. <https://doi.org/10.1161/CIRCGEN.124.004681>
28. Collins FS, Varmus H. A new initiative on precision medicine. *N Engl J Med*. 2015;372(9):793–795. <https://doi.org/10.1056/NEJMp1500523>
29. Kaufman DJ, Baker R, Milner LC, Devaney S, Hudson KL. A survey of U.S. adults' opinions about conduct of a nationwide Precision Medicine Initiative(R) cohort study of genes and environment. *PLoS One*. 2016;11(8):e0160461. <https://doi.org/10.1371/journal.pone.0160461>
30. Mora S, Wenger NK, Cook NR, et al. Evaluation of the pooled cohort risk equations for cardiovascular risk prediction in a multiethnic cohort from the Women's Health Initiative. *JAMA Intern Med*. 2018;178(9):1231–1240. <https://doi.org/10.1001/jamainternmed.2018.2875>
31. Emdin CA, Khera AV, Natarajan P, et al. Evaluation of the pooled cohort equations for prediction of cardiovascular risk in a contemporary prospective cohort. *Am J Cardiol*. 2017;119(6):881–885. <https://doi.org/10.1016/j.amjcard.2016.11.042>
32. Bycroft C, Freeman C, Petkova D, et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature*. 2018;562(7726):203–209. <https://doi.org/10.1038/s41586-018-0579-z>
33. Goff DC Jr, Lloyd-Jones DM, Bennett G, et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2014;63(25 Pt B):2935–2959. <https://doi.org/10.1016/j.jacc.2013.11.005>
34. Arnett DK, Blumenthal RS, Albert MA, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2019;74(10):1376–1414. <https://doi.org/10.1016/j.jacc.2019.03.009>
35. Levey AS, Stevens LA, Schmid CH, et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med*. 2009;150(9):604–612. <https://doi.org/10.7326/0003-4819-150-9-200905050-00006>
36. Lloyd-Jones DM, Huffman MD, Karmali KN, et al. Estimating longitudinal risks and benefits from cardiovascular preventive therapies among Medicare patients: the Million Hearts Longitudinal ASCVD Risk Assessment Tool. A special report from the American Heart Association and American College of Cardiology. *J Am Coll Cardiol*. 2017;69(12):1617–1636. <https://doi.org/10.1016/j.jacc.2016.10.018>
37. Paruchuri K, Bhukar R, Urbut S, et al. Using sex-specific polygenic risk to prognosticate coronary artery disease in women. *J Am Heart Assoc*. 2024;13(12):e034946. <https://doi.org/10.1161/JAHA.123.034946>
38. Surakka I, Wolford BN, Ritchie SC, et al. Sex-specific survival bias and interaction modeling in coronary artery disease risk prediction. *Circ Genom Precis Med*. 2023;16(1):e003542. <https://doi.org/10.1161/CIRCGEN.121.003542>
39. Oliva M, Munoz-Aguirre M, Kim-Hellmuth S, et al. The impact of sex on gene expression across human tissues. *Science*. 2020;369(6509). <https://doi.org/10.1126/science.aba3066>
40. Trinder M, Francis GA, Brunham LR. Association of monogenic vs polygenic hypercholesterolemia with risk of atherosclerotic cardiovascular disease. *JAMA Cardiol*. 2020;5(4):390–399. <https://doi.org/10.1001/jamacardio.2019.5954>
41. Wu H, Forgetta V, Zhou S, Bhatnagar SR, Pare G, Richards JB. Polygenic risk score for low-density lipoprotein cholesterol is associated with risk of ischemic heart disease and enriches for individuals with familial hypercholesterolemia. *Circ Genom Precis Med*. 2021;14(1):e003106. <https://doi.org/10.1161/CIRCGEN.120.003106>
42. Ference BA, Braunwald E, Catapano AL. The LDL cumulative exposure hypothesis: evidence and practical applications. *Nat Rev Cardiol*. 2024;21(10):701–716. <https://doi.org/10.1038/s41569-024-01039-5>
43. Khera AV, Won HH, Peloso GM, et al. Diagnostic yield and clinical utility of sequencing familial hypercholesterolemia genes in patients with severe hypercholesterolemia. *J Am Coll Cardiol*. 2016;67(22):2578–2589. <https://doi.org/10.1016/j.jacc.2016.03.520>
44. Aragam KG, Jiang T, Goel A, et al. Discovery and systematic characterization of risk variants and genes for coronary artery disease in over a million participants. *Nat Genet*. 2022;54(12):1803–1815. <https://doi.org/10.1038/s41588-022-01233-6>
45. Khera AV, Emdin CA, Drake I, et al. Genetic risk, adherence to a healthy lifestyle, and coronary disease. *N Engl J Med*. 2016;375(24):2349–2358. <https://doi.org/10.1056/NEJMoa1605086>
46. Marston NA, Kamanu FK, Nordio F, et al. Predicting benefit from evolocumab therapy in patients with atherosclerotic disease using a genetic risk score: results from the FOURIER trial. *Circulation*. 2020;141(8):616–623. <https://doi.org/10.1161/CIRCGEN.119.043805>
47. Natarajan P, Young R, Stitzel NO, et al. Polygenic risk score identifies subgroup with higher burden of atherosclerosis and greater relative benefit from statin therapy in the primary prevention setting. *Circulation*. 2017;135(22):2091–2101. <https://doi.org/10.1161/CIRCGEN.116.024436>
48. Nissen SE, Wang Q, Nicholls SJ, et al. Zetrasiran—a small-interfering RNA targeting lipoprotein(a): a phase 2 randomized clinical trial. *JAMA*. 2024;332(23):1992–2002. <https://doi.org/10.1001/jama.2024.21957>
49. O'Donoghue ML, Rosenson RS, Gencer B, et al. Small interfering RNA to reduce

lipoprotein(a) in cardiovascular disease. *N Engl J Med*. 2022;387(20):1855-1864. <https://doi.org/10.1056/NEJMoa2211023>

50. Ray KK, Wright RS, Kallend D, et al. Two phase 3 trials of inclisiran in patients with elevated LDL cholesterol. *N Engl J Med*. 2020;382(16):1507-1519. <https://doi.org/10.1056/NEJMoa1912387>

51. Nissen SE, Linnebjerg H, Shen X, et al. Lep-odisiran, an extended-duration short interfering

RNA targeting lipoprotein(a): a randomized dose-ascending clinical trial. *JAMA*. 2023;330(21):2075-2083. <https://doi.org/10.1001/jama.2023.21835>

52. Fry A, Littlejohns TJ, Sudlow C, et al. Comparison of sociodemographic and health-related characteristics of UK Biobank participants with those of the general population. *Am J Epidemiol*. 2017;186(9):1026-1034. <https://doi.org/10.1093/aje/kwx246>

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APPENDIX For supplemental figures and tables, please see the online version of this paper.